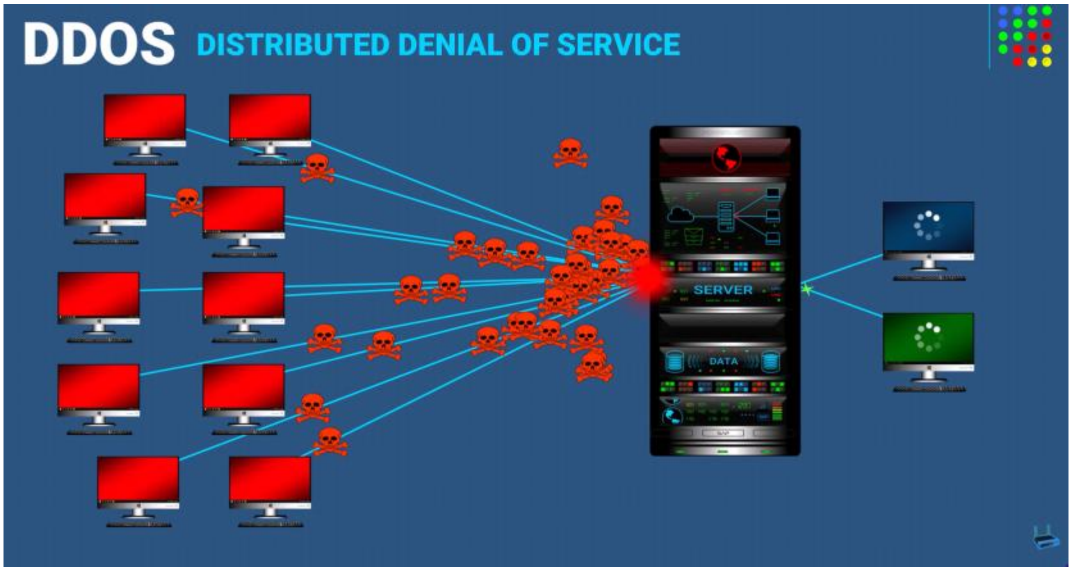
DDOS

Tuesday, May 5, 2026 4:20 PM

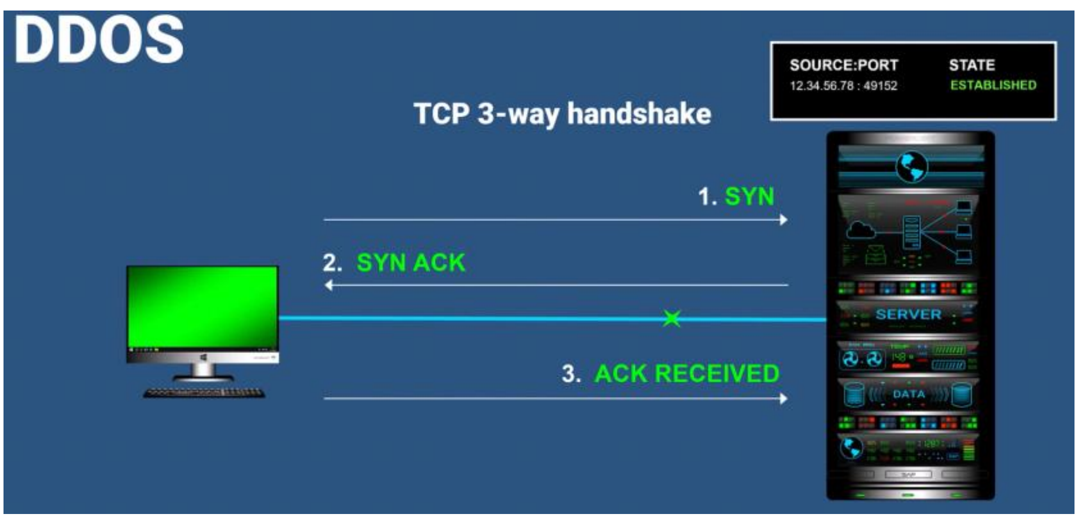
DDoS Attack Overview

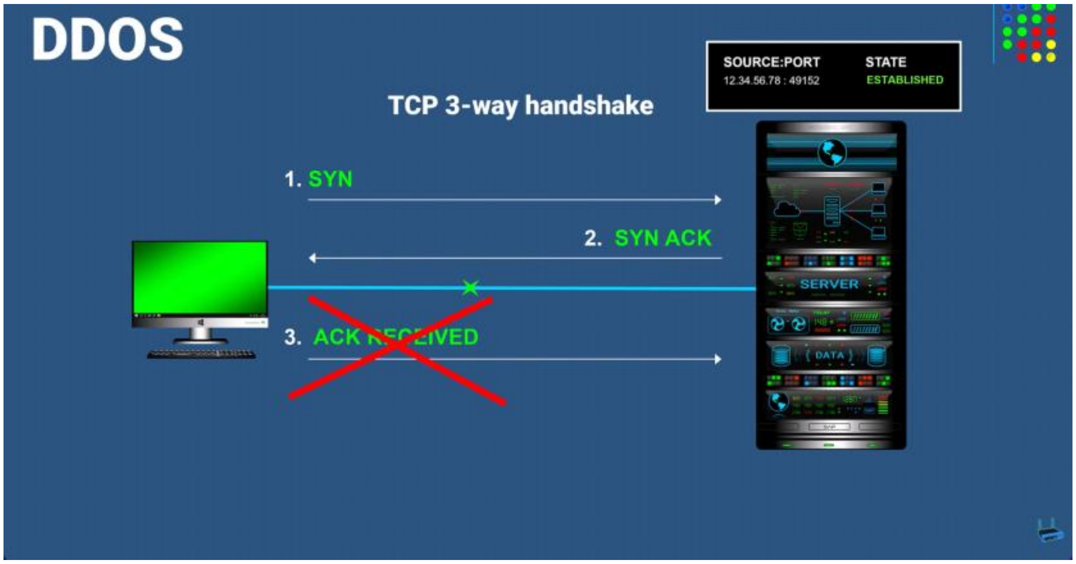
A DDoS (Distributed Denial of Service) attack is a cyber-attack designed to disrupt a server or network by overwhelming it with a massive flood of traffic. Unlike a standard DoS attack, which originates from a single source, a DDoS attack utilizes a botnet—a network of thousands of malware-infected devices controlled by a single attacker—to target a system simultaneously.



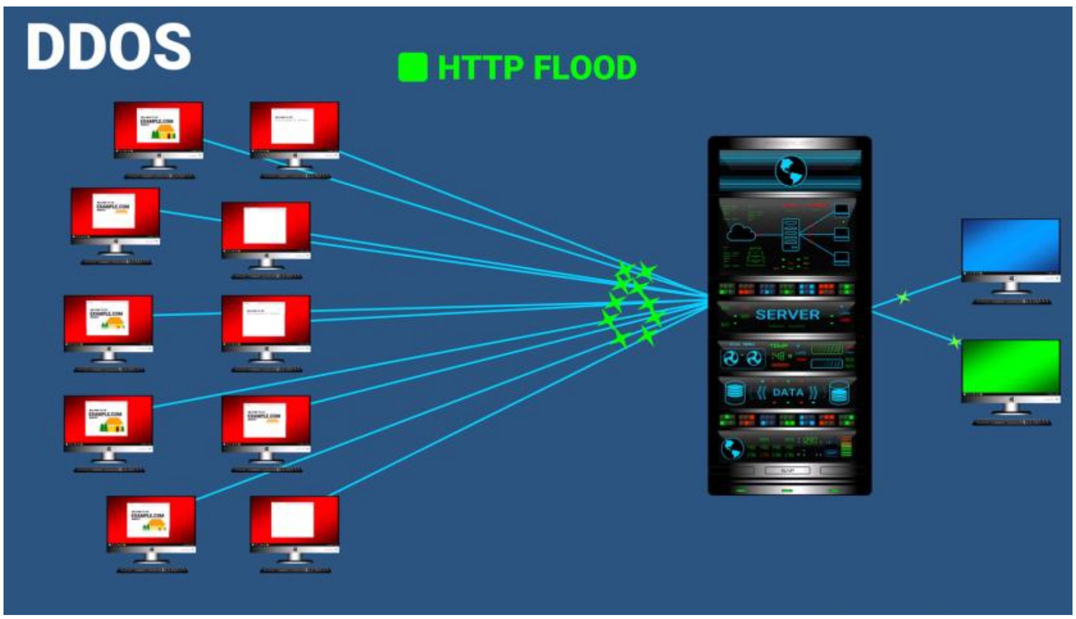
Common Attack Methods

• SYN Flood: This exploits the TCP three-way handshake. The attacker sends multiple connection requests (SYN) but deliberately fails to complete the final step of the handshake. This leaves the server with thousands of "half-open" connections, exhausting its resources.





• HTTP Flood: Multiple devices repeatedly refresh a website, flooding it with HTTP requests until the web server crashes or b ecomes extremely slow.



Identification and Prevention

Signs of an attack often include sudden, significant slowdowns, the inability to connect to a service, or server errors.

Defense strategies include:

• Rate Limiting: Restricting the number of requests a single IP address can make within a specific timeframe.
• Web Application Firewalls (WAF): Filtering incoming traffic to distinguish between malicious requests and legitimate user a ctivity.
• Load Balancers: Distributing incoming traffic across multiple servers to prevent any single point from being overwhelmed.
Motivations

Attackers typically conduct these operations for financial gain (targeting competitors), political motives, or simply for mal icious entertainment.